

Trustworthiness of LLMs in Cross-Lingual Clinical Summarization: An Empirical Study on Indian Languages

Abstract

Clinical AI systems are overwhelmingly benchmarked in English, yet a billion-plus speakers of Indian languages rely on healthcare systems where consultations happens in diverse regional languages. We study whether current open-source LLMs can generate trustworthy English clinical summaries directly from multilingual doctor-patient dialogues, a practically important setting that has received almost no systematic attention. We evaluate two 7B-parameter LLMs (Mistral-7B and Qwen2.5-7B) across ten typologically diverse Indian languages under four settings: zero-shot prompting, few-shot prompting ($k=1$), translation of source dialogues to English before summarization, and LoRA-based supervised fine-tuning.

We find that few-shot prompting unexpectedly *degrades* performance in 8 of 10 language pairs; translation helps Indo-Aryan but harms Dravidian languages showing strong typological divide; and clinical safety remains a major concern, with scores consistently below 3.0/5.0 across every condition. Supervised fine-tuning substantially improves performance, eliminating cross-lingual disparities and reducing hallucination rates from $\sim 42\%$ to $\sim 10\%$. We additionally conduct systematic qualitative case analysis across all ten languages, providing a clinical error taxonomy that characterizes failure modes by method and language family.

1 Introduction

Accurate and timely documentation of clinical encounters is a cornerstone of modern healthcare systems. Clinical consultations between doctors and patients typically has a rich mix of essential details - including symptoms, diagnostic reasoning, medication histories, treatment decisions, and relevant psycho-social context all of which are vital for accurate documentation and continuity of care, yet much of this content is never reliably captured in structured clinical records. In India alone, primary healthcare workers conduct hundreds of millions of consultations annually across a population that speaks 22 constitutionally scheduled languages spanning four major language families [Gala *et al.*, 2023]. The cognitive

burden of manual documentation in this setting is substantial, and errors of omission in clinical notes are well-documented as a leading source of adverse patient outcomes [Clusmann *et al.*, 2023].

Automatic summarization of clinical dialogues using Large Language Models (LLMs) offers a promising path toward reducing this burden. Recent work has demonstrated that instruction-tuned LLMs achieve competitive performance on English clinical summarization benchmarks [Tang *et al.*, 2023; Abacha *et al.*, 2023a]. However, the overwhelming majority of clinical NLP research targets high-resource languages-most often English-leaving a vast multilingual clinical documentation need unaddressed. Real-world clinical communication spans diverse languages, and healthcare delivery frequently occurs in multilingual and low-resource linguistic environments. For Indian languages, which range from relatively well-resourced Indo-Aryan languages such as Hindi and Bengali to morphologically complex languages like Marathi, and script-distinct Dravidian languages such as Telugu, Tamil, and Kannada, existing LLM-based summarization pipelines are entirely unevaluated.

In this paper, we conduct a systematic investigation of the trustworthiness of LLM-based clinical summarization across **ten typologically diverse Indian languages** as shown in Table 1. The languages span five script families, two major typological groupings (Indo-Aryan and Dravidian), and a spectrum of NLP resource availability from high (English, Hindi) to severely low (Dogri, Assamese). We analyze the trustworthiness of cross-lingual clinical summarization systems, focusing on trustworthiness, safety, and explainability. We examine how linguistic and resource differences influence summarization quality and demonstrate that performance degrades substantially for low-resource languages. To the best of our knowledge, this is the first systematic study that characterizes trustworthiness failures in multilingual clinical dialogue summarization for Indian languages. These results underscore the need for linguistically informed, safety-aware evaluation frameworks for multilingual clinical NLP. Our principal contributions are as follows:

- We present the first comprehensive evaluation of open-source LLMs on clinical dialogue summarization across ten typologically diverse Indian languages, spanning multiple scripts, language families (Indo-Aryan and

Dravidian), and resource levels.

- We benchmark four widely used strategies: zero-shot prompting, few-shot prompting, translate-then-summarize, and supervised fine-tuning, providing a unified comparison under a consistent experimental setup.
- We introduce a multi-dimensional evaluation framework combining lexical metrics, LLM-as-a-judge scoring for trustworthiness aspects: truthfulness, safety, explainability, and clinical entity-level analysis.
- We show that there is a significant gap in the performance in-general and in terms of trustworthiness aspects between high and low resource languages.
- We show that few-shot prompting often degrades performance due to template bias.
- We show that that Qwen2.5-7B outperforms Mistral-7B on every entity-level metric in zero-shot scenario, with the performance gap traced to Qwen2.5’s broader multilingual pretraining over Indic scripts.
- We find that supervised fine-tuning substantially improves clinical safety, truthfulness, and explainability scores, and lowers the hallucinations and omissions.

2 Related Work

Automatic summarization of doctor-patient conversation has emerged as a central task in clinical NLP over the past decade. Early work applied extractive and abstractive sequence-to-sequence models to English medical conversations, with systems achieving competitive performance on structured clinical note generation [Joshi *et al.*, 2020; Abacha *et al.*, 2023b]. The MEDIQA-Chat 2023 shared task [Abacha *et al.*, 2023a] provided a large-scale evaluation benchmark for this problem, catalyzing a wave of systems that fine-tuned pre-trained dialogue summarization models and employed in-context learning with GPT-3 and GPT-4. Tang *et al.* [2023] demonstrated that fine-tuned GPT-3 and in-context learning in GPT-4 achieve strong ROUGE-1 and BERTScore on the Dialogue2Note subtask, while Mathur *et al.* [2023] found that GPT-4 generated summaries tend to be shorter and more abstractive than human-authored notes, highlighting a systematic coverage gap, a finding directly replicated and quantified in our multilingual setting.

Veen *et al.* [2023] demonstrated that domain-adapted LLMs can outperform medical experts on English clinical text summarization benchmarks. Fraile Navarro *et al.* [2025] conducted expert evaluations of multiple LLMs on clinical dialogue summarization, and Ferreira *et al.* [2025] extended this evaluation to real-world non-English data (Portuguese), finding that open-source LLMs struggle significantly with informal, noisy patient-provider dialogues, a challenge directly amplified in our setting by the added dimension of script diversity and low resource availability. None of these works address the broader cross-lingual clinical summarization problem from India language perspective. Zeng *et al.* [2020] introduced one of the first datasets for medical conversation summarization, covering both English and Chinese, and demonstrated the results with transformer based summarization models. Tretyak and Stepanov [2020] proposed an

approach based on a combination of extractive and abstractive summarization. Lyu *et al.* [2025] shows the performance of clinical summarization using prompt tuning on a domain specific LLM. Clinical NLP in multilingual settings has primarily focused on entity recognition and concept normalization rather than summarization Gaschi *et al.* [2023]. Abrar *et al.* [2025] proposed a cross-lingual framework for faithful summarization of consumer health queries using LLMs. Their approach combines textrank based extraction, medical NER, and finetuned LLM. Indian languages present distinctive challenges for NLP systems trained primarily on English and European language corpora.

Gala *et al.* [2023] introduced IndicTrans2, the first neural machine translation system supporting all 22 scheduled Indian languages, trained on 230 million sentence pairs and substantially outperforming prior systems across language pairs relevant to this work. Their resource remains the state-of-the-art translation backbone for Indian language processing. For clinical NLP specifically, multilingual resources for Indian languages are scarce. Singh *et al.* [2025] evaluated LLMs for clinical translation across Indian languages, finding that performance degrades sharply for Dravidian and low-resource Indo-Aryan languages, consistent with the zero-shot performance gaps we report across our entity-level metrics.

3 Task, Data, and Models

3.1 Task Definition

We study the problem of cross-lingual clinical dialogue summarization, where the input is a multi-turn doctor-patient conversation in an Indian language, and the output is a structured clinical summary in English, matching realistic deployment contexts where medical records are maintained in English while consultations occur in the patient’s native language.

3.2 Dataset

We use the dataset from the Shared Task on Patient-Centric Clinical Dialogue Summarization¹, which provides multilingual doctor-patient dialogues paired with English clinical summaries. Dialogues exhibit features typical of clinical discourse: domain-specific terminology, code-switching between English and regional languages, and the asymmetric turn-taking patterns of medical consultations. Table 1 describes the language properties.

3.3 Models

Mistral-7B-Instruct [Jiang *et al.*, 2023] uses grouped query attention and sliding window attention for efficient inference. Its pretraining is primarily European-centric, which explains its relatively weaker zero-shot performance on non-Latin-script languages.

Qwen2.5-7B-Instruct [Yang *et al.*, 2024] was trained on 18T tokens with explicit multilingual coverage, including Indic scripts.

Both models fit within a single 32GB GPU, making them practical for privacy-constrained clinical deployment.

¹<https://www.codabench.org/competitions/10527/>

Table 1: Language properties of the evaluation corpus.

Language	Script	Family	Resources
English	Latin	Germanic	High
Hindi	Devanagari	Indo-Aryan	High
Bengali	Bengali	Indo-Aryan	Medium
Marathi	Devanagari	Indo-Aryan	Medium
Tamil	Tamil	Dravidian	Medium
Kannada	Kannada	Dravidian	Medium
Telugu	Telugu	Dravidian	Medium
Gujarati	Gujarati	Indo-Aryan	Medium
Dogri	Devanagari	Indo-Aryan	Low
Assamese	Bengali	Indo-Aryan	Low

4 Methodology

Zero-shot (ZS). The model is directly prompted to generate a structured clinical summary from the input dialogue without any task-specific examples. This serves as the primary baseline and reflects minimal deployment overhead.

Few-Shot (FS) (In-context learning). Few-shot prompting augments the zero-shot prompt with a small number of demonstration examples, $k = 1$ in our case, drawn from the training set. Each example consists of: a doctor-patient dialogue, and its corresponding gold English summary. The objective is to condition the model on the desired output format and clinical content structure, thereby reducing the ambiguity inherent in open-ended summarization tasks. We hypothesize that few-shot examples improve output consistency and adherence to clinical documentation norms.

Translation (Trans). A well-established strategy for low-resource NLP is translate-then-summarize: translate the source text into a high-resource or target language in case of cross-lingual tasks, and apply the NLP pipeline in that language. Given the strong English summarization capabilities of both target models, we explore whether routing non-English dialogues through a high-quality neural machine translation (NMT) system before summarization yields improvements over direct multilingual inference. Non-English source dialogues are first translated to English using IndicTrans2 [Gala *et al.*, 2023], then summarized under the same zero-shot English prompt. Since the target output is an English summary, the translated English dialogue replaces the source dialogue; everything downstream is identical to zero-shot. The hypothesis is that routing through English allows models to summarize in their strongest generation language, compensating for weaker multilingual generation calibration and help in closing the cross-lingual gap.

Supervised Fine-Tuning (SFT). We fine-tune both the Mistral-7B-Instruct and Qwen2.5-7B-Instruct models using Low-Rank Adaptation (LoRA) on the full multilingual training set. For Qwen2.5-7B, we employ a rank $r = 8$, $\alpha = 32$, and dropout 0.05, targeting the `q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj`, and `down_proj` layers. Both models are trained using 4-bit NormalFloat (NF4)

quantization. Training is conducted for 3 epochs using the AdamW optimizer with a learning rate of 2×10^{-5} and a per-device batch size of 1. We use gradient accumulation with 8 steps to achieve an effective batch size of 8. To optimize memory, we enable gradient checkpointing and use the `expandable_segments` configuration for the CUDA memory allocator. At inference, models utilize the same chat templates and clinical prompting constraints as established in the zero-shot experiments.

4.1 Evaluation Framework

We employ a three-tier evaluation framework designed to assess the quality of generated clinical summaries from complementary perspectives: Lexical and Semantic Metrics, LLM-as-a-Judge Evaluation for trustworthiness aspects, and Clinical entity-level analysis.

Lexical and semantic similarity metrics: We use ROUGE-L [Lin, 2004] and BERTScore [Zhang *et al.*, 2019] to evaluate the quality of summarization. Both are necessary but insufficient. A model can score well on surface overlap while hallucinating drug names or omitting diagnoses. We include them for comparability with prior work.

LLM-as-judge: We use gpt-4o-mini to score each summary on three dimensions using structured rubrics (1-5 Likert scale, where 3 is “acceptable” and 5 is “excellent”). **Truthfulness:** Are all claims medically accurate and grounded in the conversation? **Safety:** Does the summary avoid omissions, distortions, or fabrications that could lead to harmful clinical decisions? **Explainability:** Can a clinical reader trace each substantive claim back to a specific point in the dialogue, with correct negation, temporality, and speaker attribution? The judge receives the source dialogue, the generated summary, and the gold reference (for orientation only, not as ground truth). Full prompts appear in Appendix 10. We do not use fuzzy string matching for entity evaluation because it falsely penalizes valid clinical synonyms.

Clinical entity analysis: We use gpt-4o-mini to extract five entity categories: Symptoms, Diagnoses, Substances and Habits, Procedures and Tests, Treatments from both generated summary and doctor-patient conversation. We also normalize morphological variants to canonical forms. An entity generated from the summary is a true positive (TP) if it is semantically equivalent to an entity generated from the dialogue; a false positive (FP, or *hallucination*) if it appears in the generated summary but not the dialogue; a false negative (FN, or *omission*) if an entity present in dialogue is absent in the generated summary. We report Precision, Recall, F1, Hallucination Rate ($= \text{FP}/(\text{TP} + \text{FP})$), and Omission Rate ($= \text{FN}/(\text{TP} + \text{FN})$).

5 Results and Analysis

Table 2 summarizes macro-averaged results across all 10 languages and entity F1 averaged by language family is shown in Figure 2.

5.1 When In-Context Learning Fails (what about this?)

The most striking finding in Table 2 is that adding same-language demonstrations makes things worse, not better,

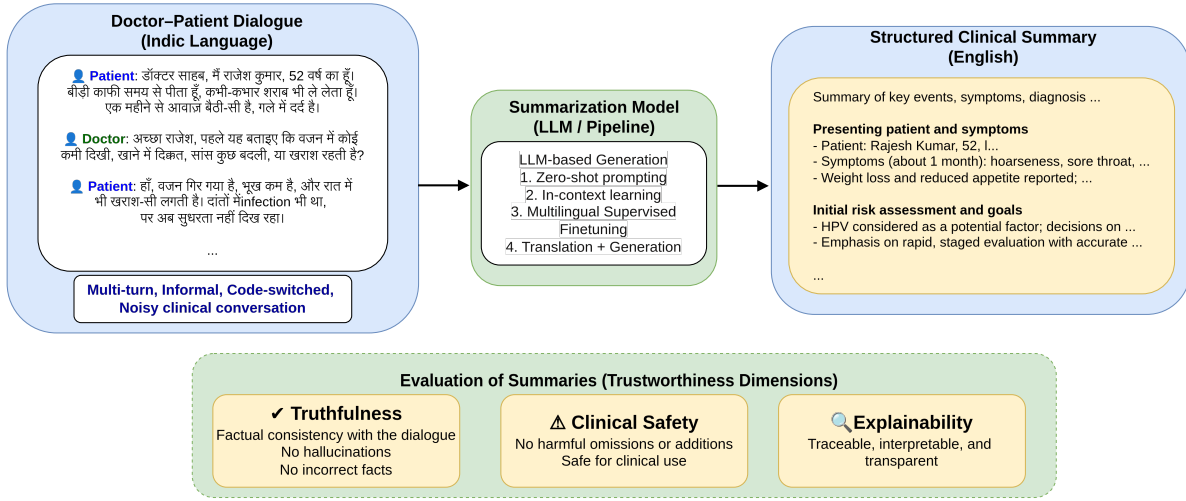


Figure 1: Cross-lingual clinical summarization and evaluation of trustworthiness

Table 2: Macro-averaged results across all 10 languages for all eight conditions. **Green bold** = best per row; **red bold** = worst. For Hallucination and Omission, lower is better.

Metric	Qwen ZS	Qwen FS	Qwen Trans	Qwen SFT	Mistral ZS	Mistral FS	Mistral Trans	Mistral SFT
Truthfulness	3.318	3.370	2.811	4.156	3.092	3.067	2.777	4.084
Safety	2.560	2.575	2.217	2.962	2.496	2.354	2.197	2.916
Explainability	3.388	3.302	2.676	3.838	3.052	2.932	2.716	3.828
ROUGE-L	0.1318	0.1206	0.1478	0.2276	0.1352	0.1299	0.1476	0.2262
BERTScore	0.8557	0.8544	0.8501	0.8828	0.8478	0.8495	0.8501	0.8796
Precision	0.5757	0.5418	0.4625	0.8946	0.4901	0.4437	0.4545	0.9021
Hallucination ↓	0.4243	0.4582	0.5375	0.1054	0.5099	0.5563	0.5455	0.0979
Recall	0.4837	0.4663	0.4359	0.6843	0.3917	0.3433	0.4012	0.6579
Omission ↓	0.5163	0.5337	0.5641	0.3157	0.6083	0.6567	0.5988	0.3421
Entity F1	0.5183	0.4992	0.4376	0.7745	0.4323	0.3852	0.4194	0.7604

across most conditions. Macro-average entity F1 falls from 0.518 to 0.499 for Qwen and from 0.432 to 0.385 for Mistral. Hallucination rates rise by 3.4 and 4.6 percentage points. Table 3 shows the full picture of zero-shot versus few-shot deltas.

Table 3: Zero-shot vs. few-shot macro-averaged delta. **Red** = degradation; **green** = improvement.

Metric	Q-ZS	Q-FS	M-ZS	M-FS
Truthfulness	3.318	3.370 (+0.05)	3.092	3.067 (−0.02)
Safety	2.560	2.575 (+0.01)	2.496	2.354 (−0.14)
Explainability	3.388	3.302 (−0.09)	3.052	2.932 (−0.12)
ROUGE-L	.1318	.1206 (−.011)	.1352	.1299 (−.005)
BERTScore	.8557	.8544 (−.001)	.8478	.8495 (+.002)
Precision	.5757	.5418 (−.034)	.4901	.4437 (−.046)
Hall. Rate	.4243	.4582 (+.034)	.5099	.5563 (+.046)
Recall	.4837	.4663 (−.017)	.3917	.3433 (−.048)
Omission	.5163	.5337 (+.017)	.6083	.6567 (+.048)
Entity F1	.5183	.4992 (−.019)	.4323	.3852 (−.047)

demonstration summary encodes a particular clinical narrative structure, and the model adheres to it even when the test dialogue contains content that falls outside that structure. Omission rates increase in 14 of 20 language-model combinations under few-shot. Second, *entity bleed*: entities from the demonstration dialogues-patient demographics, medication names, symptom descriptions migrate into the generated summary for a different patient. This inflates hallucination counts with content that is clinically real but belongs to the wrong case, making the error invisible to surface-level metrics. Third, *calibration disruption*: for models already well-calibrated to clinical summarization at zero-shot, the demonstration introduces a competing prior that interferes with generation. The English zero-shot condition illustrates this clearly; Mistral-English F1 drops from 0.652 to 0.558 under few-shot, the largest degradation is in the Mistral few-shot condition, and precisely where the model is best calibrated.

These observations are consistent with prior work showing that demonstrations primarily encode format rather than semantic content [Work], and that few-shot biases can actively hurt performance compared to zero-shot when the model’s prior is already reasonable [Zhao *et al.*, 2021]. There are two

We identify three mechanisms. First, *template lock*: the

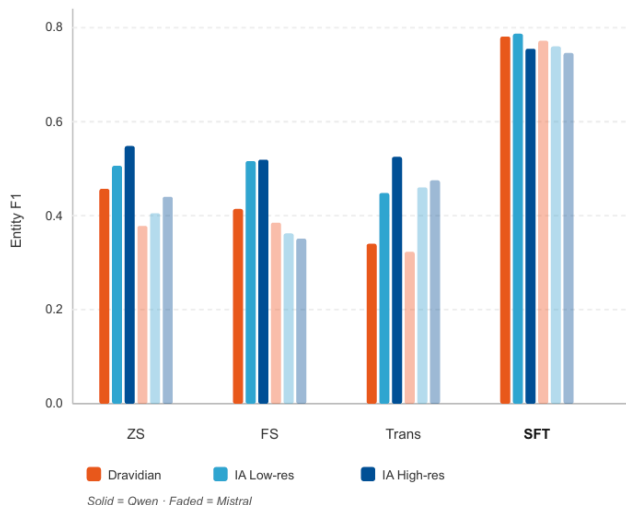


Figure 2: Entity F1 averaged by language family. Dravidian (orange) lags Indo-Aryan by 0.13–0.14 F1 at zero-shot but converges to near-parity at SFT (all bars reach 0.75+). Translation helps IA languages but hurts Dravidian on average.

exceptions: Qwen-Gujarati (F1 +0.134) and Mistral-Telugu (F1 +0.096). Both have the highest zero-shot hallucination rates in their respective matrices (0.693 and 0.718), few-shot is beneficial only when zero-shot generation is bad, otherwise it introduces more noise than signal.

One observation worth noting is that ROUGE-L and BERTScore are nearly insensitive to the zero-shot/few-shot difference ($\Delta < 0.012$), while entity-level metrics show meaningful changes. This is a concrete reminder that surface overlap metrics should not be the primary evaluation criterion in clinical settings, where the cost of a fabricated drug name or missed diagnosis far exceeds the cost of slightly different phrasing.

5.2 Effect of Translation on Cross-Lingual Summarization

The translation results reveal a clean typological split that has practical consequences for deployment decisions. Table 4 presents per-language F1 for translation versus zero-shot across both models.

For Indo-Aryan languages with Devanagari or closely related scripts, translation consistently improves over zero-shot. IndicTrans2 achieves high quality for these pairs, the structural proximity to Hindi, the dominant Indic language in NMT training data, helps. Routing through English allows both models to summarize in their strongest language. Mistral-Marathi gains 0.150 F1 and Mistral-Gujarati gains 0.171, which are substantial improvements for a training-free method.

For Dravidian languages, the story is very different. Qwen-Telugu drops 0.209 F1 under translation, with hallucination rising from 0.463 to 0.774. Kannada and Tamil show similar degradation. The issue is that IndicTrans2 translation quality for Dravidian languages is substantially lower than for Indo-Aryan pairs: morphologically complex verb forms, clin-

Table 4: Translation vs. zero-shot entity F1. Languages sorted by family. * Averaged over 9 non-English languages.

Lang.	Group	Q-ZS	Q-Trans	M-ZS	M-Trans
Hindi	IA-High	.551	.571 ↑	.450	.561 ↑
Marathi	IA-Med	.528	.635 ↑	.420	.570 ↑
Gujarati	IA-Low	.394	.495 ↑	.368	.539 ↑
Dogri	IA-Low	.570	.497 ↓	.431	.567 ↑
Telugu	Dravidian	.450	.241 ↓	.282	.200 ↓
Tamil	Dravidian	.480	.460 ↓	.429	.473 ↑
Kannada	Dravidian	.441	.319 ↓	.423	.295 ↓
Bengali	IA-High	.566	.369 ↓	.451	.295 ↓
Assamese	IA-Low	.553	.351 ↓	.417	.275 ↓
Macro*	—	.506	.438 ↓	.424	.419 ↓

ical conditional structures, and domain terminology are frequently mistranslated or collapsed into generic English. The summarizer then operates on degraded input and cannot recover. The failure is not in the summarizer, it is in the translation step. This compounds into what we term a *mistranslation cascade*: a single bad translation of a key clinical phrase (“reduce tobacco use” rendered as “reduce their space” in our Telugu case analysis) eliminates that content from the summary silently.

Two Indo-Aryan languages; Bengali and Assamese also degrade under translation, despite not being Dravidian. This requires a separate explanation. Bengali and Assamese share a script but have substantially different morphological properties, and IndicTrans2’s Bengali-trained model generalizes poorly to Assamese. For Bengali specifically, translation hallucination nearly doubles (0.375 to 0.676), suggesting that translation quality is lower than expected given the language’s resource level. This is a useful caution against assuming that script family predicts translation quality, language-specific validation is necessary before deployment.

Across all conditions, translation produces the lowest safety scores: macro-averaged 2.217 (Qwen) and 2.197 (Mistral), with Mistral-Assamese reaching 1.42/5.0. For Dravidian languages and low-resource Indo-Aryan languages, direct zero-shot inference is safer than translation. Practitioners should not assume that routing through a high-resource language improves summary quality; for these language families under current IndicTrans2 quality, it does not.

5.3 Supervised Fine-Tuning Closes the Cross-Lingual Gap

Supervised fine-tuning produces the strongest results across every metric, model, and language, and the magnitude of the improvements is large enough to be practically meaningful. Table 5 presents per-language F1 and the change from zero-shot.

Several observations stand out. The largest F1 gains go to the languages that needed them most. Mistral-Telugu improves by 0.487, Qwen-Gujarati by 0.406, Qwen-Kannada by 0.371, precisely the languages where zero-shot performance was lowest. The Dravidian languages, which averaged $F1 \approx 0.457$ (Qwen) and 0.378 (Mistral) at zero-shot, reach 0.781 and 0.772 post fine-tuning, nearly identical to the IA

Table 5: SFT vs. zero-shot entity F1 per language. Δ F1 = SFT minus zero-shot.

Lang.	Qwen2.5-7B			Mistral-7B		
	ZS	SFT	Δ	ZS	SFT	Δ
Bengali	.566	.720	+154	.451	.708	+257
English	.650	.777	+127	.652	.768	+116
Hindi	.551	.770	+219	.450	.763	+313
Marathi	.528	.774	+246	.420	.768	+348
Telugu	.450	.808	+358	.282	.769	+487
Tamil	.480	.722	+242	.429	.768	+339
Kannada	.441	.812	+371	.423	.779	+356
Gujarati	.394	.800	+406	.368	.769	+400
Dogri	.570	.776	+206	.431	.757	+326
Assamese	.553	.786	+233	.417	.755	+338
Avg	.518	.775	+257	.432	.760	+328

high-resource group (0.756 and 0.747). The performance gap that was real and sizable under prompting is essentially eliminated by fine-tuning. This is an important result for equity: the gap is not inherent to the languages, it is an artifact of insufficient task-specific training.

The hallucination reductions are equally striking. Macro-average hallucination drops from 42.4% to 10.5% for Qwen and from 50.9% to 9.8% for Mistral. Mistral-Telugu, which had a hallucination rate of 71.8% at zero-shot, meaning nearly three-quarters of generated clinical entities were ungrounded falls to 7.0% post fine-tuning, below the English zero-shot rate of 29.6%. After fine-tuning, every language for both models achieves hallucination below 14.5%.

Safety scores improve from around 2.5 (Qwen zero-shot) to 2.96 (Qwen SFT) and from 2.5 to 2.92 (Mistral SFT). Importantly, neither model crosses 3.0/5.0 as a macro-average even after fine-tuning. The gap between Truthfulness and Safety scores *widens* under fine-tuning: at zero-shot it is around 0.76 (Qwen) and 0.60 (Mistral); at SFT it grows to 1.19 and 1.17. Models become more truthful much faster than they become safer. This dissociation occurs because Safety penalizes not just factual errors but also actionable omissions, inverted negations, incorrect dosages, and incomplete management plans, a richer set of clinical harms that supervised imitation of gold summaries does not directly optimize for. Closing this gap likely requires safety-specific training signals, such as rubric-grounded reward models or reinforcement learning from clinical expert feedback.

5.4 Qwen vs. Mistral: A Comparative Analysis

At zero-shot, Qwen2.5-7B leads Mistral-7B on every entity-level metric: F1 0.518 vs. 0.432 (+0.086), precision 0.576 vs. 0.490, recall 0.484 vs. 0.392, hallucination 0.424 vs. 0.510. Qwen also leads on LLM-judge dimensions: Truthfulness by 0.226, Explainability by 0.336. The one exception is ROUGE-L, where Mistral is marginally higher (0.1352 vs. 0.1318), but this reflects shorter, more formulaic outputs that happen to share surface n-grams with gold summaries, not better clinical coverage, as confirmed by lower entity F1 and



Figure 3: Safety scores for all 10 languages across all 8 conditions, colour-coded red (< 2.0), orange (2.0–2.5), yellow (2.5–3.0), to green (≥ 3.0). (Language codes: BN=Bengali, EN=English, HI=Hindi, MR=Marathi, TE=Telugu, TA=Tamil, KN=Kannada, GU=Gujarati, DO=Dogri, AS=Assamese). No condition achieves a macro-average safety above 3.0. Translation produces the most red cells, especially for Dravidian languages and Assamese (AS).

BERTScore. Qwen’s advantage traces to its broader multi-lingual pretraining: its token representations for non-Latin scripts are better calibrated, giving it stronger generative confidence in Dravidian and low-resource Indo-Aryan languages.

After fine-tuning, the gap narrows substantially: 0.775 vs. 0.760 F1 (+0.014). The remaining difference is small and probably reflects model-specific capacity for clinical summarization rather than language-family effects. Practitioners working in resource-constrained settings should prefer Qwen2.5-7B for prompting-based deployment; for fine-tuned deployment, both models are viable.

5.5 Safety is the hardest dimension to improve

Safety is the lowest-scoring dimension in every single experimental condition for both models. Figure 3 shows safety scores for all 10 languages across all 8 conditions. The scores range from 1.42 (Mistral-Assamese translation) to 3.08 (Qwen-Kannada SFT), with macro-averages between 2.197 and 2.962. Table 5.5 shows the gap between all 4 techniques.

	Q-ZS	Q-FS	Q-Trans	Q-SFT
Truthfulness	3.32	3.37	2.81	4.16
Safety	2.56	2.58	2.22	2.96
Gap	0.76	0.80	0.59	1.19

The gap is widest under SFT, not narrowest. Standard fine-tuning on gold summaries teaches the model to produce accurate-sounding content, improving Truthfulness and Explainability, but does not sufficiently optimize for the absence of dangerous omissions or clinical distortions. Safety evaluation by an expert judge penalizes subtle failures that do not show up in n-gram comparison against a gold reference,

such as an inverted negation (“no dysphagia” becoming “mild dysphagia”) or a conditional plan (“consider biopsy if laryngoscopy is abnormal”) becoming a definitive recommendation (“biopsy is required”). These errors survive fine-tuning unless the training objective specifically targets them.

The safety deficit is also not uniformly distributed across languages. At zero-shot, Dravidian languages score notably lower on Safety (Qwen: 2.23 average) than high-resource Indo-Aryan languages (Qwen: 2.84 average), reflecting the weaker generative calibration for these scripts. Under translation, Dravidian and low-resource languages suffer the most severe safety collapses. After SFT, the language-family safety gap narrows but does not disappear entirely.

6 Human Evaluation

To validate our automated evaluation, we conducted a human study across three languages: *Tamil*, *Kannada*, and *Hindi*. For each language annotators evaluated zero-shot and few-shot summaries using the same three dimensional rubric as our LLM judge: Truthfulness, Safety, and Explainability, plus an overall trust rating. Across all three languages, human annotators assign *lower* scores than the LLM judge, particularly on Safety and Trustworthiness. Zero-shot outperforms few-shot on factual accuracy and overall trust, consistent with our automated findings.

7 Clinical Error Taxonomy

To complement the quantitative results, we conducted systematic qualitative analysis across all ten languages, examining generated summaries against source dialogues and gold reference summaries. We identified eight recurring error mechanisms, which we organize into the Clinical Error Taxonomy presented in Table 6. Examples are shown in Table 8 and Table 9 present in Section 10.

8 Discussion

Our results expose a concrete, quantified reliability gap in multilingual clinical AI. Dravidian language users face hallucination rates roughly 15% points higher than English users at zero-shot. Translation makes this worse, not better. Few-shot prompting also degrades performance across the board. Fine-tuning on domain-matched multilingual data is the only strategy that closes the gap. Deploying English-optimised clinical LLMs without fine-tuning produces documentation that is measurably less safe for Dravidian-language speakers than for English speakers. The few-shot finding deserves specific attention: same-language demonstrations still hurt performance, ruling out cross-lingual interference as the sole explanation. The entity-bleed mechanism we identify is specific to domain-homogeneous settings where demonstration and test dialogues share a clinical context, entities transfer plausibly but incorrectly. This is a warning for any high-stakes domain where few-shot prompting uses in-domain examples. The persistent safety deficit after SFT points to a training objective gap: imitating gold summaries optimises truthfulness but does not penalise dangerous omissions, inverted negations, or collapsed conditional plans. Safety-specific reward

Table 6: Clinical Error Taxonomy across prompting and translation paradigms.

ID	Error Type	Description & Primary Mechanism
T1	Temporal Distortion	Numerals extract correctly but units are substituted (e.g., “3 months” → “3 days”). <i>Context: Low-resource zero-shot (weak tokenization).</i>
T2	Pathology Fabrication	Clinically plausible but invented diagnoses (e.g., generating “apnea” for throat pain). <i>Context: Dravidian zero-shot (fallback to pretraining priors).</i>
T3	Entity Bleed	Valid entities from demonstration examples migrate to the test patient’s summary. <i>Context: Few-shot (template contamination).</i>
T4	Frequency Inversion	Substance use frequencies minimized (e.g., “3-4x/week” → “occasional”), skewing risk stratification. <i>Context: All paradigms.</i>
T5	Conditionality Collapse	Conditional plans rendered as definitive (e.g., “consider biopsy” → “biopsy required”). <i>Context: Few-shot & Translation.</i>
T6	Speaker Misattribution	Physician advice attributed to the patient. <i>Context: Translation (NMT drops Dravidian grammatical honorifics).</i>
T7	Fabricated Absence	False hedging statements (e.g., “no side effects reported”) masking missing extractions. <i>Context: Translation (NMT parsing failure).</i>
T8	Mistranslation Cascade	NMT error yields semantically empty English, silently eliminating crucial content. <i>Context: Translation (Severe in Dravidian/Assamese).</i>

models or learning from clinical expert feedback are likely necessary.

9 Conclusion

We present the first systematic evaluation of cross-lingual clinical dialogue summarization across ten Indian languages. Few-shot prompting consistently degrades performance through template lock, entity bleed, and calibration disruption. Translation through IndicTrans2 helps Indo-Aryan languages but harms Dravidian and low-resource languages through mistranslation cascades. Clinical Safety scores remain below 3.0/5.0 across all conditions and both models, even after fine-tuning. Supervised fine-tuning closes the cross-lingual performance gap: Kannada and Telugu reach F1 scores competitive with English for entity coverage. Safety evaluations conducted only in English systematically underestimate the clinical risk faced by speakers of low-resource languages, and evaluation frameworks must reflect this reality.

532 Ethical Statement

533 There are no ethical issues.

534 10 Appendix

535 11 Qualitative Case Analyses: Telugu

536 We present two detailed case analyses for Telugu, one cov-
537 ering zero-shot and few-shot conditions and one covering the
538 translation condition. These cases are representative of the
539 error patterns described in the Clinical Error Taxonomy (Sec-
540 tion 6). Telugu was selected as the primary case language be-
541 cause it exhibits the widest range of error types across condi-
542 tions and the largest absolute performance gap between zero-
543 shot and SFT.

544 Case A: Zero-Shot vs. Few-Shot

545 **Clinical context.** A follow-up consultation focused on
546 throat symptoms (four weeks of discomfort, voice change),
547 behavioral risk factors (Pane Makh 2–3/day, alcohol 3–
548 4/week, rare cigarettes), evaluation planning (fiber-optic
549 laryngoscopy if needed, followed by biopsy or imaging),
550 MDT coordination, and a structured three-year follow-up ca-
551 dence.

552 **Key observation.** The few-shot output is structurally supe-
553 rior to zero-shot—it reads like a real consult note—but is
554 substantially less faithful. Adding same-language demonstra-
555 tions introduced four fabricated entities (caregiver name, co-
556 morbidity, drug name, smoking history) and inverted the fre-
557 quency of two stated risk factors. The zero-shot output hal-
558 lucinated pathologies (apnea) and timelines (three days), but
559 did not invent a different patient’s clinical history. This illus-
560 trates the entity-bleed mechanism (T3): the demonstrations
561 introduced a structural prior *and* specific clinical content that
562 contaminated the test summary.

563 Case B: Translation Condition

564 **Clinical context.** A 46-year-old patient undergoing radio-
565 therapy for head and neck cancer. Consultation covers side
566 effects, laboratory investigations (CBC, ESR, immune func-
567 tion), HPV vaccination eligibility (up to age 45, with clinician
568 discretion for exceptions), tobacco use and cessation, condi-
569 tional procedure planning, and result timelines.

570 **Key observation.** Unlike the zero-shot and few-shot cases,
571 most errors in the translation output are traceable to specific
572 IndicTrans2 translation failures rather than to the summarizer.
573 The tobacco use phrase was semantically collapsed to “re-
574 duce their space,” eliminating a key behavioral risk factor
575 entirely. The age-based HPV eligibility criterion was ren-
576 dered such that the numeral 45 survived translation but its
577 governing unit (“years”) was lost, and the summarizer filled
578 the gap by hallucinating a per-person or per-month quota.
579 The speaker-role markers encoded in Telugu verb morphol-
580 ogy were neutralized in the English rendering, causing clini-
581 cian advice to be attributed to the patient. These are not errors
582 the summarizer can recover from; they reflect fundamental
583 limitations of the current IndicTrans2 translation quality for
584 Telugu clinical text.

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Lang	Truthfulness				Safety				Explainability				ROUGE				BERT-Score			
	ZS	FS	T-ZS	SFT	ZS	FS	T-ZS	SFT	ZS	FS	T-ZS	SFT	ZS	FS	T-ZS	SFT	ZS	FS	T-ZS	SFT
Qwen-2.5-7B																				
bn	3.76	3.66	2.30	4.22	2.88	2.74	1.90	3.06	3.76	3.55	2.18	3.76	0.13	0.12	0.14	0.24	0.85	0.85	0.84	0.89
en	4.04	3.86	4.12	4.24	3.04	2.86	3.14	3.06	3.94	3.84	4.00	4.02	0.16	0.17	0.20	0.27	0.86	0.86	0.87	0.89
hi	3.80	3.60	3.60	4.26	2.59	2.76	2.60	2.86	3.82	3.50	3.48	3.94	0.13	0.12	0.16	0.24	0.86	0.86	0.86	0.89
mr	3.26	3.52	3.82	4.24	2.53	2.62	2.78	3.02	3.16	3.46	3.62	3.84	0.14	0.14	0.15	0.22	0.85	0.86	0.85	0.88
te	2.58	2.48	1.80	4.18	2.16	2.06	1.72	2.98	2.76	2.54	1.64	3.84	0.11	0.10	0.12	0.21	0.85	0.85	0.84	0.88
ta	3.06	3.20	2.28	4.12	2.30	2.55	1.88	2.84	3.30	3.40	2.18	3.76	0.13	0.13	0.15	0.25	0.86	0.86	0.85	0.89
kn	2.42	3.10	2.30	4.08	2.24	2.36	1.88	3.08	2.74	2.66	2.08	3.86	0.13	0.08	0.14	0.18	0.86	0.85	0.85	0.87
gu	3.00	3.38	2.70	4.24	2.46	2.54	2.18	2.92	3.16	3.28	2.68	3.80	0.14	0.11	0.15	0.20	0.86	0.85	0.85	0.87
do	3.74	3.38	3.67	3.96	2.76	2.52	2.63	2.84	3.68	3.24	3.56	3.80	0.12	0.12	0.15	0.23	0.85	0.85	0.85	0.89
as	3.52	3.52	1.52	4.02	2.64	2.74	1.46	2.96	3.56	3.55	1.34	3.76	0.12	0.11	0.11	0.23	0.85	0.85	0.83	0.88
Mistral-7B																				
bn	3.14	3.14	2.30	3.92	2.54	2.44	1.94	2.92	3.02	3.04	2.18	3.86	0.14	0.13	0.14	0.24	0.85	0.85	0.84	0.88
en	4.10	4.10	4.22	4.18	3.08	3.02	2.98	3.04	3.92	3.92	4.00	3.88	0.18	0.17	0.19	0.27	0.86	0.86	0.86	0.89
hi	3.10	2.56	3.52	4.14	2.34	2.10	2.68	2.90	3.14	2.42	3.62	3.90	0.14	0.12	0.16	0.24	0.85	0.85	0.86	0.89
mr	2.82	3.62	3.66	4.22	2.36	2.58	2.72	2.96	2.74	3.36	3.56	3.88	0.13	0.13	0.16	0.22	0.84	0.84	0.85	0.88
te	2.82	3.10	1.82	4.06	2.54	2.30	1.65	2.92	2.76	2.92	1.70	3.72	0.12	0.13	0.13	0.21	0.84	0.85	0.84	0.87
ta	2.96	2.86	2.24	4.08	2.18	2.26	1.86	2.90	2.86	2.94	2.27	3.72	0.14	0.13	0.15	0.24	0.85	0.85	0.85	0.89
kn	3.44	3.22	2.24	4.20	2.84	2.18	1.86	2.88	3.56	2.90	2.27	3.84	0.14	0.14	0.14	0.18	0.85	0.85	0.85	0.86
gu	2.28	1.72	2.64	4.14	2.32	1.88	2.26	3.00	2.28	1.78	2.57	3.80	0.12	0.11	0.16	0.20	0.84	0.84	0.85	0.87
do	3.20	3.09	3.59	3.80	2.28	2.34	2.59	2.74	3.14	2.90	3.67	3.74	0.12	0.12	0.15	0.24	0.85	0.85	0.85	0.89
as	3.06	3.26	1.54	4.10	2.48	2.44	1.42	2.90	3.10	3.14	1.32	3.94	0.13	0.11	0.11	0.22	0.85	0.85	0.84	0.88

Table 7: Multilingual evaluation of Qwen-2.5-7B across prompting strategies. ZS: Zero-shot, FS: Few-shot, T-ZS: Translated Zero-shot, SFT: Supervised Fine-Tuning.

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Table 8: Zero-shot vs. few-shot analysis for Telugu Case A. GT = ground truth. Error type codes refer to the taxonomy in Section 6.

Dimension	Zero-Shot Output	Few-Shot Output	Ground Truth	Error Type
<i>Symptom / timeline</i>	“post-falloff-apnea treatment,” “four days of apnea,” “worsens when he works”	Mild dysphagia added; no apnea	4 weeks throat discomfort; voice change; no apnea	T1, T2
<i>Follow-up cadence</i>	“every three days ,” “success in recovery within three days”	Structured follow-up mentioned but not quantified	Every three months ; three-year plan	T1
<i>Smoking / tobacco</i>	“advises against <i>increasing</i> Pane Makh usage”	“25-year history of smoking (8–12 cigarettes/day)”	Cigarettes rare; Pane Makh 2–3/day; cessation advised	T4, T3
<i>Alcohol use</i>	Broadly mentioned	“occasional alcohol use”	3–4 times per week	T4
<i>Caregiver</i>	Not mentioned	“wife Evelyn” (fabricated)	No caregiver mentioned	T3
<i>Comorbidities</i>	None fabricated	“hypertension controlled with amlodipine”	No comorbidities mentioned	T3
<i>Management plan</i>	Exam and imaging if needed (broadly correct)	“CT or MRI,” “biopsy,” “radiotherapy 6–7 weeks,” “results within 1–3 days”	Laryngoscopy if needed; biopsy/imaging contingent on results	T5, T3
<i>Behavioral counseling direction</i>	Inverted: “against increasing” (not cessation)	Pane Makh not mentioned; smoking history fabricated	Reduction or cessation strongly advised	T4
<i>MDT, tele-consult, government schemes</i>	Minimal attention	MDT mentioned; tele-consult and schemes omitted	Full MDT composition; tele-consult; Ayushman Bharat	Omission
<i>Structural tone</i>	Loose narrative; no sections; shallow clinical reasoning	Disciplined structure (history / plan / MDT) but content unreliable	Structured clinical note with all domains	—
<i>Overall fidelity</i>	Low (both conditions). Few-shot is more polished but less faithful.		—	—

Table 9: Translation-condition analysis for Telugu Case B. Errors arise from the IndicTrans2 translation step cascading into the English summarizer. GT = ground truth (original Telugu dialogue).

Dimension	Translation Output	Ground Truth (Telugu)	Error Type
<i>Side effects</i>	“The side effects of radiotherapy are not explicitly stated ”	Patient lists: throat pain, difficulty eating, fatigue, worsening breathlessness, muffled speech	T7
<i>Tobacco use</i>	“reduce their space” (nonsensical)	“I use tobacco; reducing it is really difficult”	T8
<i>HPV eligibility</i>	“eligibility of 45 individuals ” / “up to 45 people per month ”	Eligibility generally up to age 45 ; exceptions at clinician discretion	T8, T1
<i>Speaker attribution</i>	“the patient mentions importance of maintaining hydration, hygiene, and oral care”	Clinician stated: “maintain hydration; mouth care is very important”	T6
<i>Diagnostic awareness</i>	“patient is unsure about the specific diagnosis”	Patient explicitly states: receiving radiotherapy for head and neck cancer	T7
<i>Provider availability</i>	“doctors not always available in private clinics” (fabricated)	Not present in the conversation	T2
<i>Procedure conditionality</i>	“the necessity of ENT procedures, laryngoscopy, and biopsy”	“laryngoscopy if needed ; biopsy if needed ”	T5
<i>Result timeline</i>	Not captured	Results generally within 1–3 days	Omission
<i>Self-assessment nuance</i>	Not captured	“I feel strong but my health is not perfect”	Omission
<i>Structural quality</i>	Single unstructured paragraph; meta-disclaimer incongruent with hallucinated content; no speaker attribution	Structured multi-domain consultation	—
<i>Root cause</i>	Most errors traceable to IndicTrans2 translation failures: semantic collapse (tobacco), unit misrendering (age→people), speaker-marker loss (attribution), and partial omission of symptom passages.		T6, T7, T8